Alzheimer’s Choice Theory

Single-Origin Closed Causal Topological Model

Mathematical Formalization of Alzheimer's Choice Theory (ACT)

A Companion Document to the Chinese Original

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Information on the Theoretical Report

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Executive Summary

Formal Translation — No Alteration to Logical Skeleton or Causal Structure.

This document constitutes a formal representation layer of the original Chinese ACT text and does not represent an independent theoretical version.

Introduction Abstract

This document presents the Mathematical Formalization (Version 1.1) of *Alzheimer’s Choice Theory (ACT)*, originally authored in Chinese by Cheng-Chun Yen (顏誠均, C.C. Yen).

The present manuscript does not introduce a new theoretical version. Rather, it provides a symbolic and structural rearticulation of the original Chinese text, preserving its logical skeleton, causal topology, and ontological commitments without modification.

ACT is defined by a single-origin closed causal topology, in which the long-term Non-Thinking State (NT) constitutes the unique endogenous generative condition leading to Neural Functional Decline (D), and subsequently to Alzheimer’s Disease (AD). This formalization translates the original conceptual framework into mathematical expressions in order to enhance structural clarity, logical examinability, and internal consistency.

All mathematical representations herein — including axioms, definitions, symbolic relations, and boundary constraints — function solely as formal encoding tools. They neither extend nor revise the underlying theory. In the event of any discrepancy between this document and the Chinese original (*Alzheimers-Choice-Theory-V1.1-Final-Draft-ZH*), the Chinese text retains sole semantic authority.

This document serves as a companion representational layer and should be read in conjunction with the original Chinese manuscript.

Authoritative Source Declaration & Statement on Mathematical Formalization

This mathematical manuscript constitutes a symbolic and formal rearticulation of the Chinese-language original: *Alzheimer's Choice Theory: Single-Origin Closed Causal Topological Model, Version 1.1 Final Draft*, authored by C.C. Yen. The technical organization of this document is derived entirely from the original manuscript.

1. No logical skeleton has been altered. No causal direction has been changed. No additional theoretical assumptions, causal relations, or model layers have been introduced beyond those established in the original text.

2. All mathematical representations — including equations, symbols, functional forms, axioms, definitions, and boundary conditions — serve solely as equivalent formal expressions of concepts already articulated in the original work, with the aim of enhancing structural clarity and logical examinability. Such formulations should be understood as translational renderings of the original conceptual structure, rather than as extensions, modifications, or augmentations of the underlying theory.

3. Mathematics herein functions as a formal expression tool, not as the theory itself. The specific mathematical formulations presented represent the currently most structurally consistent expressions under the closed-causal constraints of ACT. Should the author identify more appropriate formal expressions in the future, they may be adopted in subsequent versions. Readers who identify potentially more suitable formal renderings are welcome to participate in academic discussion.

4. In the event of any discrepancy between this mathematical formalization and the Chinese original, the Chinese original shall prevail as the sole and final authoritative interpretation.

5. This version number (V1.1) follows the numbering of the Chinese original.

Meta-Theoretical Foundations and Structural Invariance Declarations

(a) Semantic Sovereignty Convention (Meta-Theoretical Rule)

ACT is defined only in the Chinese original text: *Alzheimer's Choice Theory (V1.1, Chinese Original)*.
This document is a representational encoding only.

Formally:

- Interp(ACT_math) = Interp(ACT_Chinese)
- Authority(ACT_Chinese) > Authority(ACT_math)

where Authority(·) denotes interpretative authority (semantic sovereignty), not a mathematical order.

If any discrepancy arises among:

- symbolic expression,
- logical encoding,
- or mathematical structure,

then ACT_Chinese prevails as the sole authoritative interpretation.
Therefore, no new semantics, causation, or ontology may be inferred from mathematical abstraction alone.

(b) Topology Lock Declaration (Structural Invariance Clause)

This mathematical formalization is a representational layer of
Alzheimer's Choice Theory (ACT), V1.1 (Chinese Original).

It does not introduce new theoretical claims.
All semantic authority belongs to the Chinese original text.

Within ACT, the causal topology is strictly fixed as:

$$NT \to_c D \to_c AD$$

where:

- NT denotes the endogenous state of long-term non-thinking (長期不思考狀態),
- D denotes neural functional decline (神經功能下降),

- AD denotes Alzheimer's disease (阿兹海默症).

This topology is:

1. Closed — no additional external causal nodes may generate AD or D.
2. Non-statistical — it does not represent probabilistic association.
3. Non-capacity-based — it does not describe cognitive reserve, neural compensation, or resilience capacity.
4. Endogenous-only — the causal origin of AD is defined solely within the NT state. No exogenous or operational-layer variable may be interpreted as a moderator that alters the generative origin, causal direction, or node-dependency hierarchy of $NT \to_c D \to_c AD$.

AMT and AMT-CV-2PEM provide an operational identification rule for NT:

$$NT(x) \Leftrightarrow \Delta(x) > \varepsilon$$

where $\Delta(x) = AMTI(x) - MTSI(x)$.

This identification rule does not constitute an alternative etiology.
It does not redefine AD as a capacity depletion model.

Any interpretation that maps ACT into the framework of Cognitive Reserve, Brain Maintenance, Compensation, or Resilience models constitutes semantic misclassification.

The causal topology of ACT is invariant and non-reorderable.

(c) Non-Isomorphism Declaration

The causal topology of ACT is not structurally isomorphic to:

- Cognitive Reserve models
- Brain Maintenance models
- Compensation frameworks
- Capacity depletion theories
- Resilience-based models

Formally:

$$\neg\exists f : Iso\left(ACT_{Topology}, Reserve_{Topology}\right)$$

Iso(A,B) denotes structural isomorphism that preserves causal direction, generative origin, node hierarchy, and endogenous/exogenous status.

where structural equivalence requires preservation of:

- causal direction
- generative origin
- node dependency hierarchy
- endogenous/exogenous distinction

Isomorphism requires preservation of generative origin.

$$ACT\ origin = NT$$

$$Reserve\ origin \neq NT$$

ACT defines:

$$AD \coloneqq \text{generated solely from NT}$$

Reserve models define:

$$AD \coloneqq \text{capacity failure under pathological burden}$$

These are structurally non-equivalent causal systems.

Therefore, semantic mapping between ACT and Cognitive Reserve constitutes misclassification.

All semantic authority belongs to the Chinese original (V1.1).

(d) Semantic Anchor (語意錨點)

Alzheimer's Choice Theory (ACT) refers to an invariant closed-causal conclusion: NT → D → AD. This core conclusion remains consistent across different derivation contexts.

The present version (V1.1 Final Draft) employs a functional biological mechanism postulate — Use-Dependent Maintenance (UDM) — as its derivational starting point. The earlier version (V1.0 Final Draft) had completed its derivation within the framework of Modern/Extended Evolutionary Synthesis (MS/EES).

ACT is not a sub-theory, extension, or component of any particular derivation framework. Its core causal conclusion exists independently of the derivation path. Different contexts serve only as formal choices for derivation and interpretation, not as necessary preconditions for the theory's validity.

Formally:

$$Conclusion(ACT) = (NT \to_c D \to_c AD)$$

$$\forall F \in \{UDM, MS/EES, Future\_Framework\}\colon\ Derivable(Conclusion(ACT), F) \to Valid\big(Conclusion(ACT)\big)$$

$$\neg\exists F\colon\ Conclusion(ACT) \subset F$$

where F denotes a derivation framework. The conclusion is invariant under framework substitution.

(e) Methodological Clarification (方法論聲明)

The core causal conclusion of ACT (NT → D → AD) constitutes a closed and self-sufficient causal chain whose validity is not bound to any particular derivation framework.

V1.0 Final Draft employed Modern Evolutionary Synthesis / Extended Evolutionary Synthesis as its derivational basis. V1.1 Final Draft adopts Use-Dependent Maintenance (UDM) as a functional biological mechanism postulate. This change is a technical operation aimed at tightening theoretical boundaries and simplifying the derivation chain, not a negation of the previous derivation path.

Evolutionary theory is retained in V1.1 only as historical motivation (context of discovery) and does not enter the justification chain (context of justification).

(f) Theory Type Declaration (理論類型聲明)

ACT is classified as:

- Closed-Causal Theoretical Framework
- Structural Redefinition Theory

ACT does not belong to incremental modifications of existing mainstream models, but constitutes a foundational reframing of the causal structure through redefinition of the key generative origin.

(g) Structural Author Constraint (結構性作者約束)

V1.1 Final Draft explicitly reinforces author attribution and internal theoretical perspective markers in the formulation of core definitions, propositions, and causal conclusions, in order to maintain semantic integrity and clarity of authorship.

This mechanism covers:

- Definition methods for key terminology
- Marking formats for closed-causal propositions
- Semantic boundaries of non-generic model names
- Theoretical perspective positioning of key statements and summary sentences in each section

The purpose is to reduce the risk of de-authoring and misattribution that may arise during semantic extraction, summary restatement, cross-language transcription, or secondary interpretation.

Formally:

$$\forall P \in CorePropositions(ACT):\ Attribution(P) = \text{Cheng-Chun Yen (C.C. Yen)}$$

$$\forall D \in Definitions(ACT):\ SemanticAuthority(D) = ACT_{Chinese\ Original}$$

This constraint has been incorporated into the structural principles of the main text as a fixed standard for subsequent version revisions and textual review.

Table of Contents

§1 Overall Structural Overview of ACT — 全文整體架構說明

General Structural Overview

This paper is divided into two main sections:

1. Main Theory (Foundational Theory)
2. Appendix (Theoretical Extensions)

The two sections are logically distinct:

- The Main Theory constitutes the core and foundation of ACT.
- The Appendix is an applied extension developed in response to technological developments, and is not a necessary component of the ontological causal proposition.

Main Theory (Foundational Theory):

The Main Theory consists of two logical layers and two parts:

Main Theory — Foundational Theory Part I:
Layer 1: The Foundational Layer — Alzheimer's Choice Theory Framework of Non-Action

Contents:

- Postulate: Use-Dependent Maintenance (器官功能維持依賴持續使用, UDM)
- Core Hypothesis: Non-Thinking Hypothesis (不思考假說, NTH)
- Consequence of NTH: Non-Degenerative Brain Hypothesis (大腦不退化假說, NDBH)

Functions of this layer:

- Establish the causal origin of ACT
- Define the sole endogenous causal mechanism of Alzheimer's disease
- Form the closed causal chain:

$$NT \rightarrow D \rightarrow AD$$

Layer 1 (The Foundational Layer) has the following characteristics:

1. It is the ontological core of ACT.
2. It belongs to the disease causation layer.
3. It does not depend on Layer 2 theory for its validity.

4. It does not presuppose any applied model.

In other words:

Layer 1 theory is logically self-sufficient and does not require AMT as a condition for its validity.

Main Theory — Foundational Theory Part II:
Layer 2: The Mechanism Modeling Layer

Contents:

- AMT Theory
- AMT-CV-2PEM
- AMTI / MTSI
- Autognosis Drive

Functions of this layer:

- Explanation and substantive content of NTH
- Explain "how thinking exists and is maintained"
- Establish a thinking intensity assessment mechanism
- Describe how to avoid entering long-term NT

Important declarations:

1. Layer 2 is not a justificatory premise of Layer 1.
2. Layer 1 hypotheses are not invalidated by revisions to Layer 2 models.
3. AMT is an independent methodological model.
4. Layer 2 does not possess disease-causal authority.

Therefore:

Layers 1 and 2 are mutually independent in theoretical validity.

Layer 1 defines disease causation.
Layer 2 explains the thinking mechanism.

Appendix (Theoretical Extensions)

Layer 3: The Applied Extension Layer

Contents:

- LLM-HITL-SCCLF
- SMDL

- ATP

Functions of this layer:

- Thought activation mechanism
- Thought maintenance technology
- Human-machine interaction framework

This layer has the following characteristics:

1. It belongs to technological extension.
2. It can be replaced as technology advances.
3. It does not constitute a necessary condition for the Main Theory's validity.
4. It does not possess disease-causal status.
5. It does not affect the theoretical validity of Layers 1 and 2.

Therefore:

The Appendix is an open application layer, not a closed causal layer.

Structural Hierarchy Diagram:

```
Main Theory (Foundational Theory)
│
├─ (Part I) Alzheimer's Choice Theory Framework of Non-Action — Layer 1: The
│    Foundational Layer (Disease Causal Core) NT → D → AD
│
└─ (Part II) Layer 2: The Mechanism Modeling Layer (Thinking Intensity Model)
      AMT Theory / AMT-CV-2PEM


Appendix (Theoretical Extensions)
│
└─ Layer 3: The Applied Extension Layer (Technological Extension)
      LLM-HITL-SCCLF / SMDL / ATP
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Core Logical Boundaries (Must Be Fixed)

1. Sole disease-causal layer and foundational layer: Layer 1 only.
   Main Theory: Part I.
2. Mechanism modeling layer: Layer 2. Main Theory: Part II.
3. Applied extension layer: Layer 3. Appendix.

4. The three layers are mutually independent in terms of validity conditions, but conceptually expand outward from the core.
5. Revisions, replacements, or failures of Layers 2 and 3 do not affect the validity of Layer 1 causal propositions.
6. The Appendix is replaceable; the Main Theory is not.
7. Layer 1 constitutes the sole disease-causal declaration layer of ACT; other layers do not possess disease-causal declaration authority.

Final Summary

The Main Theory constitutes the closed causal core of ACT; the Appendix is an applied extension developed in response to technological developments. Layers 1 and 2 are mutually independent in theoretical validity; Layer 3 does not constitute a necessary condition for theoretical validity.

§2 Theoretical Hierarchy and Methodological Distinction — 理論層級說明與方法論區分

In C.C. Yen's Alzheimer's Choice Theory (ACT) research framework, theoretical hierarchy and methodological hierarchy must be clearly distinguished.

This research takes the functional biological mechanism postulate "Use-Dependent Maintenance (UDM)" as its starting point, deriving the core hypothesis:

Non-Thinking Hypothesis (NTH)

with the Non-Degenerative Brain Hypothesis (NDBH) as a consequence of NTH.

Forming Layer 1 (The Foundational Layer):

$$Foundational_{Layer} = \{UDM_{(postulate)}, NTH_{(core\ hypothesis)}, NDBH_{(consequence\ of\ NTH)}\}$$

On this foundation, C.C. Yen separately constructs Layer 2 (The Mechanism Modeling Layer):

Abstract Multidimensional Thinking (AMT)

AMT-CV-2PEM (Constant–Variable Two-Phase Evaluation Model)

Forming the Layer 2 set:

$$Mechanism_{Modeling_{Layer}} = \{\text{AMT,AMT-CV-2PEM}\}$$

It must be explicitly noted:

(1) AMT and AMT-CV-2PEM do not serve as "deductive proof tools" for NDBH or NTH;

(2) They serve only as operationalized and structured criteria pathways for "thinking state/thinking intensity";

(3) The causal topology remains fixed as: NT → D → AD (Layer 1 proposition).

I . Hierarchical Structural Relationship:

$$Foundational_{Layer} = \{UDM,\ NTH,\ NDBH_{(\leftarrow NTH)}\}$$

$$Mechanism_{Modeling_{Layer}} = \{\text{AMT,AMT-CV-2PEM}\}$$

Where:

- The Foundational Layer defines the disease causal direction
- The Mechanism Modeling Layer defines the observable and measurable representation of thinking state

The two layers are different types of theoretical hierarchy, not a logical entailment relationship.

II . Methodological Distinction Principle

If the empirical results of AMT-CV-2PEM are consistent with the theoretical expectations of NDBH ∧ NTH, this can be formally expressed as:

$$Consistent(\text{AMT-CV-2PEM, NDBH}\wedge\text{NTH})$$

Then:

$$Consistent(\text{AMT-CV-2PEM, NDBH}\wedge\text{NTH}) \rightarrow IndirectSupport(\text{NDBH}\wedge\text{NTH})$$

But it must not be expressed as:

$$\text{AMT-CV-2PEM} \vdash (\text{NDBH}\wedge\text{NTH})$$

or

$$Proof(\text{NDBH}\wedge\text{NTH})$$

That is:

- The validity of the AMT model may constitute indirect support
- But does not constitute final proof

This distinction is designed to avoid theoretical circular confirmation.

III . Independence Declaration

AMT Theory and the AMT-CV-2PEM model possess structural independence.

Formally:

$$\text{AMT-CV-2PEM} \notin Foundational_{Layer}$$

$$AMT\text{-}CV\text{-}2PEM \notin \{UDM, NTH, NDBH\}$$

That is:

- Even if NTH awaits further testing, AMT Theory can still stand as an independent thinking structure model;
- Even if AMT Theory requires further empirical strengthening, this does not constitute a direct refutation of NDBH or NTH.

Hierarchical type distinction:

$$\{UDM,\ NTH,\ NDBH\} \subset Causal_{Proposition_{Layer}}$$

$$\{\text{AMT,AMT-CV-2PEM}\} \subset Mechanism_{Modeling_{Layer}}$$

IV . Functional Positioning

The design purpose of the AMT-CV-2PEM model is:

To transform the multidimensional structure of abstract thinking into observable and measurable criteria mechanisms.

Its role is defined as:

$$Role(\text{AMT-CV-2PEM}) = Operational_{Evaluation_{Framework}}$$

And:

$$Role(AMT - CV - 2PEM) \neq Causal_{Origin}$$

V . Summary Statement

In this research framework:

$$Theoretical_{Foundation} \coloneqq \left\{UDM_{(postulate)},\ NTH_{(core)},\ NDBH_{(consequence)}\right\}$$

$$Mechanism_{Modeling_{Pathway}} \coloneqq \{AMT, AMT - CV - 2PEM\}$$

The two are separated and type-independent.

This design:

- Avoids theoretical circular confirmation
- Maintains logical separation between causal propositions and empirical tools
- Preserves theoretical closure with methodological openness

$$\{UDM,\ NTH,\ NDBH\} \subset\ Causal_{Proposition_{Layer}}$$

$$\{AMT,\ AMT - CV - 2PEM\} \subset\ Mechanism_{Modeling_{Layer}}$$

This is hereby declared.

§3 ACT Foundational Theory Part I — The Foundational Layer — 第一層：理論基礎層

I . Symbol System

Let:

- AD: Alzheimer's Disease
- D: Neural Functional Decline
- NT: Non-Thinking State (長期不思考狀態) — long-term state of insufficient thinking
- T: Sustained Thinking Activity
- UDM: Use-Dependent Maintenance (器官功能維持依賴持續使用) — postulate
- FF: Functional Fact — the core function of the cerebral cortex and hippocampus is higher-order cognitive activity (thinking)
- BP: Bridging Premise — "use" of an organ means executing its core function
- C: Identifiable Endogenous Mechanism
- W: Wild Animal Population (empirical support level, not axiomatic level)
- NDBH: Non-Degenerative Brain Hypothesis (大腦不退化假說) — consequence of NTH
- NTH: Non-Thinking Hypothesis (不思考假說) — core hypothesis

Definition:

$\rightarrow c$ denotes causal generative relation.

$$A \rightarrow cB$$
$$\coloneqq \ A\ is\ the\ unique\ endogenous\ causal\ origin\ condition\ that\ generates\ B\ within\ ACT.$$

where A is "endogenous mechanism or state condition"; B is "outcome event/state".

Representational-layer definition of the Non-Thinking State (NT) (transcription only, no new theory introduced):

Within ACT, the operational identification indicator of "long-term non-thinking" is represented at Layer 2 by the gap between AMTI and MTSI.
This notation is introduced solely for identification purposes and does not constitute a

definition of NT's ontological nature or generative source.
Hence $NT_{AMT(x)}$ serves as an identification marker, not an ontological definition.

$$AMTI(x) \in \mathbb{R}^{+},\ MTSI(x) \in \mathbb{R}^{+}$$

$$\Delta(x) \coloneqq AMTI(x) - MTSI(x)$$

Let ε be the permissible gap threshold of AMT-CV-2PEM (a representational-layer parameter, not a physiological constant):

$$NT(x) :\Leftrightarrow \Delta(x) > \varepsilon$$

Note (semantic anti-infiltration): This representational layer does not use any exogenous threshold symbol (e.g., $\theta(x)$). ε represents only "the permissible gap relative to the individual's own $AMTI$" and must not be interpreted as a proxy threshold for education, social interaction, stimulation quantity, or other open-framework variables.

II . Postulate

A1 (Use-Dependent Maintenance, UDM)

$$A1\text{: } UDM$$

$$UDM \coloneqq \forall O \in Organ_{System}\text{: } LongTermAbsence\big(Use(O)\big) \\ \rightarrow FunctionalDecline(O)$$

Semantic note:
Any organ system, under long-term absence of use, is expected to exhibit observable functional decline.
This is a functional biological mechanism postulate, applicable at the organ-system level.
This postulate is not inductively derived from the success cases below, but serves as the theoretical starting point, evaluated by means of testable predictions.

III . Accommodation of Known Instances

If UDM holds, then any organ system under long-term absence of use should exhibit observable functional decline. The following are known instances consistent with this prediction:

$$Muscle:\ LongTermAbsence\big(Use(Muscle)\big) \rightarrow Sarcopenia$$

$$Bone:\ LongTermAbsence\big(WeightBearing(Bone)\big) \rightarrow Osteoporosis$$

$$Cardiopulmonary:\ LongTermAbsence\big(AerobicLoad(CP)\big) \\ \rightarrow FunctionalDecline(CP)$$

$$Joint:\ LongTermAbsence\big(Movement(Joint)\big) \rightarrow MobilityLoss(Joint)$$

The above are existing success instances of UDM's predictions, not the logical source of UDM.
UDM as a mechanism postulate derives its credibility from the breadth of its applicability and the accuracy of its predictions, not from the number of inductive samples.

IV . Functional Fact

FF (Core functional fact of the cerebral cortex and hippocampus)

$$FF \coloneqq CoreFunction(CerebralCortex \cup Hippocampus) = ThinkingActivity$$

Semantic note:
The core function of the cerebral cortex and hippocampus is higher-order cognitive activity：memory integration and retrieval, language comprehension and production, logical reasoning, decision-making and planning, spatial navigation, and the construction and manipulation of abstract concepts.
This is a descriptive fact from neuroscience, independent of the postulate and not derived from it.

Additional fact: The pathological progression of AD primarily attacks the cerebral cortex and hippocampus.

$$PrimaryTarget(AD) = CerebralCortex \cup Hippocampus$$

V . Bridging Premise

BP (Mediating premise)

$$BP \coloneqq \forall O \in Organ_{System}:\ Use(O) = Execute\big(CoreFunction(O)\big)$$

Semantic note:
For any organ, “use” as referred to here means the organ being employed to execute its core function.
The “use” of muscle is contraction and load-bearing, not being massaged or heated; the “use” of bone is weight-bearing, not being wrapped or protected.
Likewise, if the core function of the cerebral cortex is higher-order cognitive activity, then the “use” of the cerebral cortex is thinking.
If core functional activity is chronically insufficient, this constitutes “absence of use” as defined by the postulate.

VI . Derivation

From A1 (UDM), FF, and BP combined:

$$UDM \wedge FF \wedge BP$$

$$\vdash$$

$$LongTermAbsence(ThinkingActivity) \\ \rightarrow FunctionalDecline(CerebralCortex \cup Hippocampus)$$

That is:

$$NT \rightarrow D$$

Derivation notes:

(1) UDM’s scope of applicability is at the “organ system” level; the cerebral cortex and hippocampus, as part of the organ system, are not presumed to be exceptions.

(2) From BP, for the cerebral cortex and hippocampus, “use” is locked to “thinking” (Thinking Activity).

(3) From FF, the attack target of AD is precisely the above core functional system.

(4) Therefore, “Non-Thinking State (NT)” is positioned in this chapter as: the state in which core functional activity of the cerebral cortex and hippocampus has been chronically absent.

VII . Argument Type Statement

The argumentative form of this derivation is independent instantiation of the same postulate across different organ systems, not argument by analogy.

$$\neg(Analogy(Muscle, Brain) \vdash NTH)$$

$$UDM(\forall O) \wedge (Muscle \in Organ_{System}) \wedge (CerebralCortex \in Organ_{System})$$

$$\vdash UDM(Muscle) \wedge UDM(CerebralCortex) \quad independently$$

The strength of analogy depends on how similar two things are; the strength of independent instantiation depends on how broad the postulate's scope of applicability is. ACT relies on the universality of the postulate, not on the similarity between brain and muscle.

Burden of Proof Reversal:

$$\neg\exists_{documented}\ Attack(UDM, Muscle) \wedge \neg\exists_{documented}\ Attack(UDM, Bone)$$

If an objector wishes to claim that the cerebral cortex and hippocampus are the sole exception to UDM, the objector bears the burden of proof to independently explain why this cognitive system is not subject to the postulate. Until independent reasons are provided, the cerebral cortex and hippocampus, as organ systems, are presumed to be bound by UDM.

VIII . Core Hypotheses

D1 (Non-Thinking Hypothesis, NTH) — Core hypothesis

Based on the above derivation (VI), C.C. Yen proposes the core hypothesis:

$$D1\ (NTH):\ NT \rightarrow_c D \rightarrow_c AD$$

Semantic note:
Long-term insufficient thinking (NT) is the endogenous starting point of neural functional decline (D); neural functional decline in turn leads to the clinical manifestation of Alzheimer's disease (AD).

The causal direction must be explicitly marked:
It is not "decline leads to inability to think," but "not thinking leads to decline."

D2 (Non-Degenerative Brain Hypothesis, NDBH) — Consequence of NTH

$$D2\ (NDBH)\colon\ NTH \vdash NDBH$$

$$NDBH \coloneqq \neg(InherentAgingMechanism(CerebralCortex \cup Hippocampus) \\ \rightarrow FunctionalDecline)$$

Semantic note:
As a consequence of NTH (not an a priori truth), the higher-order cognitive functional decline of the cerebral cortex and hippocampus is not an inherent, inevitable aging mechanism, but rather the result of specific long-term state conditions (NT).
Under conditions where thinking activity is continuously maintained, the cognitive functions of the cerebral cortex and hippocampus should be maintainable.

Hierarchical relationship between NDBH and NTH:

$$NTH \in Core_{Hypothesis}$$

$$NDBH \in Consequence(NTH)$$

$$NDBH \notin Independent_{Axiom}$$

IX . Closed Causal Topology Declaration

$$NT\ \rightarrow_c D\ \rightarrow_c AD$$

$$\neg\exists X\ (X\ \rightarrow_c AD\ \wedge\ X\ \neq\ NT\ \wedge\ X\ \neq\ D)$$

NT is the unique endogenous starting node:

$$\neg\exists Y\ (Y\ \rightarrow_c D\ \wedge\ Y\ \neq\ NT)$$

All external conditions (education, social interaction, occupation, lifestyle, etc.) are modulators that may affect whether an individual falls into the NT state, but do not possess upstream causal sovereignty.
Aβ deposition, Tau protein tangles, and other pathological markers are downstream manifestations, not upstream causes.

X . Empirical Consistency

The following empirical observation patterns are compatible with the above theoretical constraints but do not constitute the logical source of NTH:

(1) Sustained T is associated with lower AD risk.

(2) NT is structurally associated with AD.

(3) In natural wild animal populations (W), sustained T(w) is observed ($\forall w \in W$, Sustained(T(w))), and no systematic population-level AD has been documented to date.

$$EC \coloneqq Compatible(Observations, NTH) \wedge \neg(Observations \vdash NTH)$$

Empirical constraint (not an axiom):

$$\forall x \in W,\ \neg NT(x) \quad (empirical\ structural\ constraint,\ not\ axiom)$$

Semantic note:
Within ACT's framework, the living conditions of natural wild animal populations do not allow for the stable existence of the long-term non-thinking state NT.
This is an empirical structural constraint, not an axiom.
It can be overturned by empirical counterexamples (see Falsifiability Condition 1).
The derivation of NTH does not depend on this empirical constraint — NTH is derived from the postulate (UDM) + functional fact (FF) + bridging premise (BP). This empirical observation provides support independent of the derivation process.

XI . Falsifiability Conditions

The falsifiability conditions of this theory are divided into four levels, corresponding to three different types of failure:

Condition 1: Wild Population Counterexample [Fatal to theory]

If all five of the following sub-conditions are simultaneously met:

(a) Environmental condition: Natural state or controlled environment simulating natural conditions, where individuals retain autonomous foraging, social interaction, territorial defense, and other behavioral patterns requiring sustained cognitive engagement;

(b) Species condition: The species relies on sustained cognitive activity for survival in its natural ecology;

(c) Status condition: Verified through behavioral observation or neurological indicators independent of AMT that the population is not in the NT state;

(d) Pathological condition: Full clinical disease course consistent with human AD (not merely pathological analogs such as Aβ deposition, but progressive loss of cognitive function to collapse of self-care ability);

(e) Statistical condition: Population-level, non-sporadic, reproducible proportion.

$$\exists P \subseteq W\ \left(Env(P) \land Species(P) \land \neg NT_{verified}(P) \land FullAD(P) \land PopLevel(P) \right)$$

→ If all five are met, ACT does not hold. If any condition is unmet, it does not constitute a valid counterexample.

Condition 2: Necessity Counterexample [Fatal to theory]

Under long-term longitudinal observation, after excluding short-term stimulation or compensatory behaviors, if systematically and reproducibly confirmed in an observable, non-sporadic population:

$$\exists P \left(Stable(P) \land Reproducible(P) \land PopulationLevel(P) \land \forall x \in P \left(\neg NT_{verified}(x) \land FullAD(x) \right) \right)$$

where $\neg NT_{verified}(x)$ must be confirmed through long-term thinking-state verification independent of AMT.
→ If established, the core proposition of NTH does not hold.

Condition 3: Generative Sovereignty Counterexample [Theory revision]

If an exogenous variable X exists that, without altering the NT state, can stably and reproducibly independently produce a full AD clinical disease course:

$$\exists X, P \left(Stable(P) \land Reproducible(P) \land PopulationLevel(P) \land \forall x \in P\left(X(x) \land \neg \Delta NT(x) \land FullAD(x) \right) \right)$$

→ If established, ACT's claim of "sole upstream endogenous generative condition" requires revision.

Condition 4: Model-Level Falsifiability [Kills tool, not theory]

AMT and MTSI are only operational identification tools for NT, not the ontological definition of NT.

If in population-level, reproducible long-term follow-up, the following repeatedly occurs:

(A) False Positive:

$$\exists P\left(Stable(P) \wedge Reproducible(P) \wedge PopulationLevel(P) \wedge \forall x \in P\left(NT_{AMT}(x) \wedge \neg NT_{verified}(x)\right)\right)$$

(B) False Negative:

$$\exists P\left(Stable(P) \wedge Reproducible(P) \wedge PopulationLevel(P) \wedge \forall x \in P\left(\neg NT_{AMT}(x) \wedge NT_{verified}(x)\right)\right)$$

→ AMT identification rules require revision or are invalidated. However, the revision or failure of the AMT model does not directly constitute a counterexample to NTH (the Layer 1 ontological proposition).

Hierarchical Distinction of Falsifiability Results

1. Core proposition failure (fatal falsification): Conditions 1 and 2. NT is not a necessary condition; ACT's core proposition does not hold.
2. Topological exclusivity failure (topological revision): Condition 3. NT's "single-origin closed causal topology" requires revision.
3. Instrument-level failure: Condition 4. Operational tools require revision or replacement; does not directly refute Layer 1.

※ Positioning note: Falsifiability exists primarily at the "state variable level" (presence or absence of NT state), not at the "intervention compliance level" (whether the individual complies with intervention).

XII . Closed Causal Topology Overview

UDM (Postulate)

↓

$$UDM \wedge FF \wedge BP \vdash (NT \rightarrow D)$$

↓

D1 (NTH): Core Hypothesis

$$NT \rightarrow_c D \rightarrow_c AD$$

↓

D2 (NDBH): Consequence of NTH

↓

Closed Causal Uniqueness:

$$(NT \rightarrow_c D) \wedge \neg\exists Y(Y \rightarrow_c D \wedge Y \neq NT)$$

Structural Summary

This version explicitly distinguishes:

- Postulate (UDM)
- Functional Fact (FF)
- Bridging Premise (BP)
- Core Hypothesis (NTH)
- Consequence (NDBH)
- Empirical Consistency
- Falsifiability Conditions (four levels)

The purpose of this formalization is to:

- Enhance clarity
- Mark logical hierarchy
- Preserve closed causal structure
- Ensure the derivation chain fully corresponds to the postulate-deductive path of Chinese Version V1.1

This section presents a constrained causal hypothesis derived from the Use-Dependent Maintenance postulate and functional facts.

Closed Causal Uniqueness Axiom

Within ACT's defined closed causal topology (Chinese Original V1.1), NT is the only defined endogenous generative condition that satisfies the causal requirement of Alzheimer's disease.
This uniqueness is a theoretical claim subject to falsification (see Falsifiability Conditions above), not a logical necessity.

$$(NT \ \rightarrow c\ D) \land \neg\exists Y(\ Y \ \rightarrow c\ D \ \land \ Y \ \neq \ NT\ )$$

within ACT's closed causal topology.

Therefore,

$$NT \rightarrow_c D \rightarrow_c AD.$$

As defined above, NT is the unique endogenous generator of D.

This uniqueness condition is structural, not statistical.
It excludes:

- capacity depletion models
- reserve-based buffering models
- compensatory frameworks
- probabilistic association models

AMT and AMT-CV-2PEM only provide an identification rule:

$$NT(x) \Leftrightarrow \Delta(x) > \varepsilon$$

They do not introduce alternative causal origins.

<u>Absorptive Closed Framework Declaration</u>

ACT does not subsume other theories as sub-theories, nor does it share causal sovereignty with open statistical models.

$$\neg(ACT \supset_{causal} Reserve_{Model}) \land \neg(ACT \supset_{causal} BrainMaintenance_{Model})$$

The term "absorption" refers to semantic repositioning under an invariant generative origin (NT), not causal incorporation or meta-theoretic unification.

Existing research observations (education, occupation, social activity, amyloid, tau, inflammation, vascular pathology) are repositioned as outer projections within the closed causal chain, without being admitted into the generative structure.

Semantic authority remains with the Chinese original.

§4 ACT Foundational Theory Part II — The Mechanism Modeling Layer — 第二層：理論機制建模層

I . Preface

In the preceding Foundational Layer, D1 (Non-Thinking Hypothesis, NTH) has been established as the endogenous starting point of the core causal chain. The closed causal topology is:

$$NT \ \to_c D \ \to_c AD$$

However, NTH only identifies "long-term non-thinking" as the endogenous starting point, and has not yet formally defined:

- What constitutes "thinking"
- What constitutes "long-term"
- What constitutes "sufficient"

To avoid semantic ambiguity and enhance operationalizability, the internal structure of "thinking" must be further delineated.

AMT Theory is introduced in this context as a structured explication of the internal variables of NTH.

In other words:

NTH defines the causal direction;
AMT defines the structure of thinking.

AMT does not add new causal chains; AMT only provides the operational identification rule for NT.

II . Structural Outline (NTH ↔ AMT Relationship)

NTH (Non-Thinking Hypothesis): Provides the causal direction

$$NT \ \to_c D \ \to_c AD$$

AMT / AMT-CV-2PEM: Provides the internal criterion mechanism for whether NT holds.
Through the two-phase model, determines whether an individual falls into NT.

Positioning conclusion:

AMT does not alter the existing causal chain;
AMT only provides structural criteria for NT.

III . Symbol Table (Extending the Existing Symbol System)

Building on existing symbols (AD, D, NT, T, UDM, FF, BP, C), the following AMT-block symbols are added:

Symbol	Definition
AMT	Abstract Multidimensional Thinking (抽象空間多維度思維力)
AMTI	AMT upper-bound index (Phase 1; theoretical individual endogenous ceiling)
MTSI	Current thinking state (Phase 2; variable value; thinking strength)
Δ	Gap: $\Delta = AMTI - MTSI$

IV . Property Declaration (Structure Only, Not Dynamic)

$$AMTI \in \mathbb{R}^{+}$$

AMTI is treated as the individual endogenous ceiling in the theoretical structure; it does not vary with MTSI.

$$MTSI \in \mathbb{R}^{+}$$

MTSI is a variable value.

This layer's formalization:

- Does not introduce time derivatives
- Does not define change rules
- Does not establish dynamic equations
-

V . AMT-CV-2PEM Two-Phase Assessment Model

1. Gap definition

$$\Delta = AMTI - MTSI$$

2. Thinking criterion (self-comparison principle)

Case 1: Thinking strength sufficient

$$\Delta \leq \varepsilon \Leftrightarrow \neg NT$$

Case 2: Thinking strength insufficient

$$\Delta > \varepsilon \Leftrightarrow NT$$

Connecting back to Layer 1 closed causal chain (unchanged):

$$NT \rightarrow_c D \rightarrow_c AD$$

VI . Complete Logical Articulation

$$(MTSI \approx AMTI) \rightarrow \neg NT(MTSI \ll AMTI) \rightarrow NT$$

Connecting back to Part I:

$$NT \rightarrow_c D \rightarrow_c AD$$

VII . Positioning Statement (MTSI Refinement)

AMTI: Theoretical ceiling (Phase 1)
MTSI: Current state (Phase 2)

This section defines only which variables affect the structural position of MTSI, and does not define the temporal evolution of MTSI.

VIII . Structural Representation of MTSI (Non-Dynamic)

$$MTSI \in \Phi_{M(S,I)}$$

MTSI is constrained by (S, I), but (S, I) is not a causal starting node of NT/D/AD.

Φ_M is merely a structural constraint mapping, not a dynamic function, nor a causal generative relation.

IX . Variable Set Definitions

1， Structural Variables (Macro-level)

$$S = \{S_1, S_2, S_3\}$$

Where:

- S_1: Social institutions
- S_2: Technological environment
- S_3: Local culture and spatiotemporal context

S represents macroscopic environmental conditions that individuals cannot directly control.
No direction is defined; no weights are defined.

2， Individual Variables

$$I = \{I_1, I_2, I_3, I_4, I_5\}$$

Where:

- I_1: Overall life condition (occupation-centered)
- I_2: Lifestyle habits (health-centered)
- I_3: Education and values
- I_4: Personality traits — Autognosis Drive (自發求知驅動力)
- I_5: Other — full scope of MTSI individual-level variables (technically defined thinking-range modulation variables, e.g., neurobiochemical mechanisms of the brain)

Autognosis Drive is the core construct of I_4.
I_5 is a technically open item to cover future identifiable individual-level modulation factors, and does not alter the closed causal topology.

X . Structural Positioning of MTSI

$$0 \leq MTSI \leq AMTI$$

And:

The position of MTSI is delineated by (S, I).

That is:

Under given structural conditions and individual conditions, MTSI falls within a certain feasible interval.

No weights, no linear combinations, no summation forms are defined.

XI . Connection to NTH

$$MTSI \ll AMTI \rightarrow NT$$

And:

$$MTSI \in \Phi_{M(S,\, I)}$$

MTSI is constrained by (S, I), but (S, I) is not a causal starting node of NT/D/AD.

Therefore, in structural representation, “conditional constraint” and “causal chain” must be distinguished:

(S, I) delimits the feasible interval of MTSI;

$$\Delta = AMTI - MTSI$$

$$\Delta > \varepsilon \rightarrow NT$$

$$NT \rightarrow_c D \rightarrow_c AD$$

Where:

S and I serve only as condition sets that influence MTSI,
and do not constitute direct causal starting points for NT, D, or AD.

NT remains the sole endogenous starting point of ACT’s closed causal chain.

XII . Methodological Treatment of Education

$$Education \subset I_3$$

Education is one possible variable influencing MTSI,
and must not be written as:

$$Education \rightarrow MTSI$$

to avoid falling into the structure of open-type reserve models.

XIII . Falsifiability Conditions

Model-level falsifiability: Targeting AMT/criteria models, not NTH ontology.

NT Identification Model Failure Condition (AMT Identification Failure Condition)

AMT and MTSI are only operational identification tools for NT (long-term non-thinking state), not the ontological definition of NT.
Therefore, the falsifiability of the AMT model should be based on whether its identification results are consistent with long-term thinking-state verification independent of AMT, not merely on whether AD occurs.

(A) False Positive

If in population-level, reproducible long-term follow-up, the following repeatedly occurs:

Individuals judged by AMT/MTSI to be in NT_AMT(x),
but confirmed to be ¬NT(x) after independent long-term thinking-state verification,

then AMT's operational identification rule for NT requires revision or invalidation.

Formal representation:

$$\exists P\left(Stable(P) \wedge Reproducible(P) \wedge PopulationLevel(P) \wedge \forall x \in P\big(NTAMT(x) \wedge \neg NT(x)\big)\right) \Rightarrow \frac{Revise}{RejectAMT} - criterion$$

(B) False Negative

If in population-level, reproducible long-term follow-up, the following repeatedly occurs:

Individuals judged by AMT/MTSI to be ¬NT_AMT(x),
but confirmed to be NT(x) after independent verification,

then AMT's identification rule also requires revision or invalidation.

Formal representation:

$$\exists P\left(Stable(P) \land Reproducible(P) \land PopulationLevel(P) \land \forall x \in P\big(\neg NTAMT(x) \land NT(x)\big)\right) \Rightarrow \frac{Revise}{RejectAMT} - criterion$$

Model-Ontology Hierarchical Declaration：

NT_AMT(x) is AMT's operational indicator for NT(x).
Its goal is to approximate NT(x), but the two are not necessarily equivalent, and identification error is permitted.

The revision or failure of the AMT model
does not directly constitute a counterexample to NTH (the Layer 1 ontological proposition).

The falsifiability of NTH ontology remains subject to the necessity counterexample and generative sovereignty counterexample specified in Layer 1.

XIV . Language Control Principle

This section (structural extension layer) deals only with the static structural topology of AMT and NTH, and does not involve temporal indexing or dynamic representation.

Therefore, the following concepts or representational forms are not introduced in this section:

• Rate of increase
• Diminishing function
• Convergence rate
• Dynamic equilibrium
• Stable point
• MTSI(t) or any time-indexed representation

Temporal indices (e.g., Δ_t, $MTSI_t$, $MTSI_{t+1}$) appear only in the subsequent "Dynamic Representation Tools" section as formal representational tools for semantic compression, and do not constitute theoretical premises of this layer.

All representations in this section are limited to static structure and logical criteria definitions.

XV . Current Structural Overview (Static Representation)

$$MTSI \in \Phi_{M(S,I)}\Delta = AMTI - MTSI\Delta > \varepsilon \rightarrow NTNT \rightarrow_{c} D \rightarrow_{c} AD$$

This is a completely static closed structure.

§5 Autognosis Drive (自發求知驅動力) — Structural Positioning Statement

To avoid theoretical-level confusion and ensure the structural clarity of the AMT-CV-2PEM two-phase assessment model, a definitive statement is hereby made regarding the role positioning of Autognosis Drive.

I . Original Theoretical Motivation

The theory originally intended to establish the following structural chain:

$$Genetic\ Factors \rightarrow Personality\ Disposition \rightarrow Autognosis\ Drive \rightarrow MTSI \rightarrow NT$$

This chain is logically reasonable, aiming to connect:

Genetic factors → Personality → Thinking behavior → Whether one falls into non-thinking state (NT)

However, within the formalized framework of AMT-CV-2PEM, strict role distinctions must be maintained at each layer.

II . Hierarchical Distinction Between AMTI and Autognosis Drive

(1) Positioning of AMTI

$$AMTI \in \mathbb{R}^{+}$$

AMTI is the individual's theoretical ceiling (potential capacity upper bound), treated in this structure as the individual's endogenous capacity upper bound.

AMTI represents:

The range of "where the individual can potentially reach."

(2) Positioning of Autognosis Drive

Within the individual variable set:

$$I = \{I_1, I_2, I_3, I_4, I_5\}$$

Definition:

$$I_4 = Autognosis\ Drive$$

Autognosis Drive belongs to personality disposition and motivational constructs.
Its function is to influence the interval in which MTSI resides, not to determine the AMTI ceiling.

Formally:

$$MTSI \in \Phi_{M(S,I)}$$

Not:

$$Autognosis\ Drive \Rightarrow AMTI$$

III . Principled Distinction Between Capacity and Motivation

This theory explicitly distinguishes:

$$AMTI \neq Autognosis\ Drive$$

Capacity ≠ Motivation

If Autognosis Drive were incorporated into the AMTI layer, it would cause logical ambiguity in the two-phase model and conflate “capacity ceiling” with “willingness to use.”

Therefore:

- AMTI = Capacity ceiling
- Autognosis Drive = Internal tendency to actively approach the ceiling

IV . Relationship to Genetic Factors

In theory, the following structure is permissible:

$$Genetic\ Factors \rightarrow AMTIGenetic\ Factors$$
$$\rightarrow Personality\ DistributionPersonality\ Distribution$$
$$\rightarrow Autognosis\ Drive$$

However, this model does not assert:

$$Autognosis\ Drive \rightarrow AMTI$$

Nor does it assert:

$$Autognosis\ Drive = Genetic\ Determinism$$

Autognosis Drive may be influenced by personality and genetic distributional tendencies, but its essence belongs to motivational and dispositional conditions, not to the capacity ontology.

V . Structural Relationship to MTSI

$$0 \leq MTSI \leq AMTIMTSI \ \in \ \Phi_{M(S,\, I)}$$

Where:

$$I_4 = Autognosis\ Drive$$

Autognosis Drive influences the position where MTSI resides, but does not alter the AMTI ceiling.

VI . Articulation with NTH

$$\Delta = AMTI - MTSI\Delta > \varepsilon \rightarrow NT$$

Therefore:

$$Autognosis\ Drive \ \subset \ I \ \wedge \ MTSI \\ \in \ \Phi_{M(S,I)} \wedge \ (\Delta \ > \ \varepsilon \ \rightarrow \ NT) \wedge \ (NT \ \rightarrow_c D \ \rightarrow_c AD)$$

Within this structure:

NT remains the sole endogenous starting point.
Autognosis Drive only indirectly affects the NT determination through the structural position of MTSI.

VII . Dynamic-Layer Boundary Declaration

This declaration belongs solely to the structural formalization level.

This section does not introduce:

- Time derivatives
- Dynamic gain terms
- Convergence rates
- Difference equations

For example:

$$MTSI(t+1) = MTSI(t) + \alpha \cdot Autognosis\ Drive$$

does not belong to this layer's theoretical content.

If future work enters the engineering dynamic modeling layer, Autognosis Drive may be represented as a parameter or stimulus factor, but its mathematization belongs to the Layer 2 representational tool, and does not alter the ontological causal structure.

VIII . Philosophical Boundary Summary

$AMTI =$ "Where one can potentially reach"

$Autognosis\ Drive =$ "Whether one is willing to go"

$MTSI =$ "Where one currently stands"

These three layers are strictly separated to maintain the structural stability of the AMT-CV-2PEM two-phase model.

IX . Final Structural Overview

$$\begin{gathered} Genetic\ Factors \to AMTI Genetic\ Factors \to Personality \\ \to Autognosis\ Drive MTSI \in \Phi_{M(S,\,I)} \Delta = AMTI - MTSI \Delta > \varepsilon \\ \to NTNT \ \to_c D \ \to_c AD \end{gathered}$$

The purposes of this statement are:

- Prevent conflation of capacity and motivation
- Maintain clarity of the two-phase model
- Prevent premature infiltration of dynamic layers
- Reserve space for future engineering representation layers

This constitutes the structural positioning clarification of AMT Theory.

§6 Dynamic Representation Tools — 動態表示工具章節

Tool Selection Statement

In C.C. Yen's Alzheimer's Choice Theory (ACT), MTSI is defined as "the indicator of current or recent thinking strength," a variable value that can be assessed at any time; AMTI is the individual's theoretical thinking-intensity ceiling, structurally treated as a fixed upper bound.

Based on the original text's semantic descriptions of "approaching," "diminishing effect," and "influenced by individual and structural variables," this section selects only one discrete-time representation form as a dynamic representational tool, used to compress and express existing semantic relationships.

This representation:

- Does not introduce any new causal propositions;
- Does not introduce new theoretical layers;
- Serves only as a formal transcription of original-text semantics;
- Final interpretive authority rests with the Chinese original.

Discrete-time and continuous-time forms are mathematically interconvertible. Differential representation was considered, but to avoid introducing additional assumptions such as continuous differentiability, the discrete-time form is adopted at this stage as the minimal representation. If higher-order engineering analysis is needed in the future, the representational tool may be replaced without affecting ACT's theoretical ontology.

Dynamic Ontology Limitation Clause

The discrete-time representation introduced in this section does not define a generative dynamical ontology of ACT, nor does it establish a state-transition law of the theory.

Formally:

The expression

$$MTSI_{\{t+1\}} = \Pi_{\{[0,AMTI]\}}\left(MTSI_t + \alpha K_{t(AMTI - MTSI_t)}\right)$$

shall be interpreted as a semantic compression tool that expresses conditional variation tendencies already described in the Chinese original text.

It does not:

- define an independent state-transition system,
- introduce a new causal layer,
- redefine NT as an emergent equilibrium state,
- transform (S, I) or K_t into causal origin nodes,
- modify the closed causal topology $NT \rightarrow c\ D \rightarrow c\ AD$.

NT remains structurally defined solely by:

$$\Delta_t > \varepsilon$$

independent of the discrete representation.

The dynamic expression does not generate NT.
It only provides a compact formal depiction of how MTSI may vary under structural constraints.

The closed causal origin of Alzheimer's disease remains:

$$NT \rightarrow c\ D \rightarrow c\ AD$$

and is not derived from the dynamic expression.

Rationale for the Dynamic Representation Section

The purpose of adding "dynamic representation tools" to this section is not to expand or revise the ACT theoretical ontology, but is based on two structural considerations:

First, to avoid excessive narrative elaboration in the full ACT text that might distract from the reading focus, the semantic relationships related to "variation of thinking strength (MTSI)" are consolidated here, using mathematical form as a compression tool, enabling readers to grasp the theoretical core in minimal space.

Second, through the consolidated discrete-time representation, the articulation relationship between AMT Theory (particularly the AMTI-MTSI two-phase criterion) and NTH is formally unified, so that all parts of the theory present consistent correspondence at the structural level. This consolidation is intended to provide a comparable mathematical expression framework to aid understanding, not to add new theoretical layers.

Therefore, the mathematical representation in this section serves only as formal compression and consolidation of existing semantics, with the function of enhancing the clarity and overall readability of the theoretical structure, without altering ACT's existing closed-causal claims.

Discrete Dynamic Representation

I . Basic Setup

AMTI = Individual theoretical ceiling (fixed value)
$MTSI_t$ = Thinking strength at the t-th assessment

Gap definition:

$$\Delta_t = AMTI - MTSI_t$$

II . Individual and Structural Variables (Folded Representation)

Let:

$K_t =$ folded representation of the influence intensity of the (S, I) variable set

where (S, I) includes, per the original text's semantics:

- Life condition
- Occupational type
- Values and education
- Personality traits
- Autognosis Drive

and includes the "MTSI diminishing effect" scenario described in the original text.

K_t here is merely a compressed representation of the aggregate influence intensity of (S, I) on MTSI;

it does not redefine variable classifications,
does not alter AMT's structural layering,
does not decompose weights or define mechanism functions.

Special note:

The "MTSI diminishing effect" is treated in this representation as one situational state of $\mathrm{K_t}$.
When tasks are repeated over long periods and stimulus novelty declines, $\mathrm{K_t}$ may weaken, thereby making MTSI difficult to maintain or tending toward decline.
This does not hypothesize that thinking strength possesses a natural dissipation mechanism; rather, it is a conditional stimulus-weakening phenomenon.

III . Discrete-Time Expression (Bounded Projection, Representational Tool Only)

Define the projection operator: $\Pi_{\{[0,AMTI]\}(z)} := \min\big(AMTI, \max(0,\ z)\big)$

One possible compressed representation consistent with the original semantics may be written as:

$$MTSI_{\{t+1\}} = \Pi_{\{[0,AMTI]\}}\big(MTSI_t + \alpha\, K_t(AMTI - MTSI_t)\big)$$

Where:

$\alpha \geq 0$ is a representation weight, with no commitment to it being a physiological constant;
K_t is the folded intensity representation of (S, I), with no weight decomposition or mechanism function;
$\Pi_{\{[0,AMTI]\}(\cdot)}$ is used only to ensure the representation stays within bounds, preventing the expression from being misread as newly introduced physiological dynamics.

IV . Semantic Correspondence Notes

(1) Approaching AMTI

$$\alpha K_t \Delta_t$$

This indicates that when $\Delta_t > 0$,
MTSI has a tendency to move closer to AMTI, and that tendency is influenced by individual and structural variable intensity.

(2) Diminishing Effect

No independent dissipation term is introduced in this representation.

When tasks are repeated long-term, challenges are lacking, or stimulation weakens, K_t may decline, thereby weakening the MTSI improvement effect or showing a declining tendency over multiple iterations.

The "diminishing" here is a condition-dependent phenomenon, not inherent natural decay of thinking strength.

Formal Statement (Conditional Nature of the MTSI Diminishing Effect):

$$K_t \downarrow \Leftarrow Condition(TaskRepetition, NoveltyDecline)$$

$$\neg\big(K_t \downarrow \Leftarrow InherentDissipation(ThinkingStrength)\big)$$

Semantic note:
The decline of K_t is caused by conditions such as long-term repetition of the same tasks and declining stimulus novelty. This is a technically inevitable phenomenon of brain automatization and efficiency gain (universal across all individuals, independent of occupational category).
This representation does not assume that thinking strength possesses a natural dissipation mechanism independent of conditions.
The "MTSI diminishing effect" and I_4 (Autognosis Drive) are concepts at different levels and must be distinguished.

V . Articulation with Risk Criteria

$$\Delta_t > \varepsilon \rightarrow NTNT \rightarrow_c D \rightarrow_c AD$$

This dynamic representation operates only on the variation representation of MTSI and does not alter ACT's existing closed causal chain.

Summary

This difference equation is merely a compressed representation of the following three semantic items from the original text:

- MTSI is a variable value;
- MTSI is influenced by individual and structural variables;
- Under certain conditions, a diminishing effect due to stimulus weakening may occur.

This representation does not introduce new causal mechanisms, nor does it hypothesize that thinking strength has natural dissipation properties.

This expression is the current minimal representational tool; in the future, other mathematical representations (e.g., differential forms) may be substituted, but all interpretations are ultimately governed by the Chinese original.

VI . Semantic Correspondence Table

This section explicitly lists the one-to-one correspondence between each component of the discrete-time expression and the semantics of the Chinese original, demonstrating that this expression introduces no mathematical concepts beyond those listed below.

Discrete-time expression:

$$MTSI_{t+1} = \Pi_{[0,AMTI]}\big(MTSI_t + \alpha K_t(AMTI - MTSI_t)\big)$$

Semantic correspondence:

(1) AMTI carries no time subscript (fixed value)

Original-text semantics: “constant, invariant, fixed”; “thinking capacity ceiling determined by genes”; “Phase 1 assessment.”

Balloon-in-Balloon analogy (Bobo 氣球比喻): “The outer balloon differs in size for each individual”; “the outer AMTI balloon is fixed.”

Seated Forward Bend analogy (坐姿體前彎比喻): “The maximum limit of flexibility indeed varies by individual”; “skeletal structure is the primary limiting factor… an innate difference that cannot be changed by postnatal training.”

(2) $MTSI_t$ carries a time subscript (variable value)

Original-text semantics: “variable value”; “indicator of current or recent thinking strength”; “a variable value that can be assessed at any time”; “Phase 2 assessment.”

Balloon-in-Balloon analogy: "The inner MTSI balloon"; "the only adjustable variable is the inner MTSI balloon."

Seated Forward Bend analogy: "Measured again at age 40, 50, or 60"; "comparing with oneself" — the individual's AMTI remains unchanged; MTSI at different time points is compared.

(3) $(AMTI - MTSI_t)$: Directional approaching term

Original-text semantics: "approaching, reaching, or getting close to"; "whether MTSI reaches or approaches AMTI."

Balloon-in-Balloon analogy: "The inner balloon MTSI 'inflates (thinks)' to approach the outer balloon AMTI" — the direction of approach is explicitly from MTSI toward AMTI, not arbitrary.

(4) K_t: Folded intensity representation of (S, I) variables

Original-text semantics: "MTSI is influenced by structural variables (social institutions, technological environment, local culture) and individual variables (lifestyle, habits, education/values, personality traits/Autognosis Drive, other)."

K_t is merely a compressed representation of the aggregate influence intensity these variables exert on MTSI. No weight decomposition, no mechanism functions, no variable reclassification.

(5) K_t may decline: MTSI diminishing effect

Original-text semantics: "An individual, due to long-term or high-frequency repetition of the same task/work content, experiences a decline in thinking-strength stimulation within that same content"; "habit, habitual actions and habitual thinking."

This is a condition-dependent phenomenon (technically inevitable brain automatization and efficiency gain), not inherent natural dissipation of thinking strength.

(6) $\Pi_{[0,AMTI]}(\cdot)$: Projection operator (boundary guarantee)

Original-text semantics: "MTSI can reach or approach AMTI, but cannot exceed it"; "0 ≤ MTSI ≤ AMTI."

Balloon-in-Balloon analogy: The inner balloon cannot inflate beyond the outer balloon.

(7) $\alpha \geq 0$: Representation weight

No commitment to it being a physiological constant. Required only for the formal needs of the discrete-time representation, to adjust the scale of the expression. Does not constitute a new theoretical parameter.

(8) Time subscript t: Assessment time point

Original-text semantics: "Assessment can be conducted anytime, anywhere"; "a variable value that can be assessed at any time."

t merely denotes the t-th assessment; no fixed interval or continuous time is presupposed.

Closure Statement

The above eight items exhaust all mathematical components of the discrete-time expression.
Every item has an explicit semantic source in the Chinese original.
This discrete-time expression introduces no mathematical concepts beyond those in the correspondence table above.

$$\{AMTI,\ MTSI_t,\ (AMTI - MTSI_t),\ K_t,\ K_t \downarrow,\ \Pi_{[0,AMTI]},\ \alpha,\ t\}$$
$$\leftrightarrow \{Original\ Semantic\ Set\}$$

This correspondence is bijectively complete: every dynamic semantic item in the original text is captured by the discrete-time expression; every mathematical component of the expression can be traced to the original text.

§7 Relationship Between Foundational Theory and Theoretical Extensions — 本論與附論之關係說明

Alzheimer's Choice Theory (ACT) is explicitly divided into two structural levels in its paper structure:

1. Main Theory: Foundational Theory
2. Appendix: Theoretical Extensions

I . Main Theory: Foundational Theory

The Foundational Theory layer constitutes a closed causal framework. Its core propositions, causal topology, and falsifiability conditions are logically independent of any specific technology, tool, or implementation form.

Structurally:

- Part I of the Main Theory:
  NTH and NDBH constitute the causal proposition layer.
- Part II of the Main Theory:
  AMT Theory and its model AMT-CV-2PEM constitute the operationalized criterion layer for thinking state.

Formally expressed as:

$$Defines(Foundational_{Hypotheses}, Causal_{Topology})Provides(AMT - CV - 2PEM, Operational_{Evaluation})$$

AMT and AMT-CV-2PEM may provide indirect support for the foundational hypotheses, but do not constitute final proof.
The two are logically related but belong to different hierarchical levels, to avoid self-circularity.

II . Appendix: Theoretical Extension Layer

The purpose of the Appendix is not to constitute a necessary premise of the Foundational Theory, but to address the following question:

Does the Foundational Theory possess practical and operational feasibility?

The Appendix provides only one feasible operational example under contemporary technological conditions, to illustrate how to maintain or enhance the key variables referenced by the theory in real-world systems.

Its formal positioning is:

$$Enables\left(Theoretical_{Foundation},\ Possible_{Implementation_{Layer}}\right)$$

The human-machine interaction and Large Language Models (LLM) discussed in the Appendix are merely one specific implementation pathway under the current technological context.

In theory:

- It is not the only pathway
- It is not a necessary condition
- It can be replaced or re-implemented

If more advanced or fundamentally different technological systems emerge in the future, they may replace the implementation layer shown in the Appendix without affecting the Foundational Theory's validity conditions.

III . Evaluation Principle Declaration

Support, questioning, or refutation of ACT should clearly distinguish:

$$Foundational\ Theory \neq Theoretical\ Extensions$$

If the technical implementation in the Appendix is proven inadequate, outdated, or replaceable by a better approach, that result affects only the choice at the operational layer and does not constitute a negation of the Foundational Theory.

IV . Hierarchical Relationship Summary

$$Foundational\ Theory \rightarrow Closed\ Causal\ Structure Theoretical\ Extensions \rightarrow Replaceable\ Implementation\ Layer$$

The Foundational Theory is a closed causal proposition;
the Appendix is a replaceable implementation representation.

This distinction ensures:

- The theory's closure is not affected by technological change
- The operational layer has methodological flexibility
- If future work enters engineering dynamic models or control system representations, such representations belong solely to Appendix-layer instrumental modeling and do not alter the foundational causal structure.

This is hereby declared.

§8 Appendix (Theoretical Extension) — The Applied Extension Layer — 第三層：理論實踐應用層

This Appendix belongs to ACT's Theoretical Extensions, with the purpose of exploring the practical feasibility of the Foundational Theory under contemporary technological conditions, rather than constituting a necessary premise of the Foundational Theory.

The overarching framework of the Appendix is:

Semantic-Motivational Dual Loop Model (語意—動機雙層閉環模型, SMDL)

SMDL is an operational framework illustrating how to maintain or enhance the continuity of thinking activity in real-world systems.

I . Dual-Layer Structure of SMDL

SMDL consists of two complementary but hierarchically distinct parts:

(1) Motivational Layer: Dopaminergic Brain Reward Circuit

This layer corresponds to the physiological and neurological motivation-maintenance mechanism.
Its function is to:

- Provide conditions for internal motivational reinforcement of thinking activity
- Form sustained engagement motivational feedback

This layer does not constitute a new disease cause; it pertains only to the maintenance mechanism of thinking activity.

(2) Semantic Layer: LLM-HITL-SCCLF

The semantic layer is:

LLM-HITL Semantic Constructive Closed-Loop Framework (SCCLF)

This framework uses human-machine interaction as its concrete projection form. Its functions include:

- Providing semantic challenges
- Delivering comprehension feedback
- Promoting cognitive stimulation

LLM is explicitly defined here as:

“Closed mediator of thought activation”

Its role is as an operational projection layer, and NOT:

- A thinking subject
- A cognitive substitute
- An emotional attachment object
- The ontological source of thinking motivation

II . Theoretical Positioning of ATP (Addiction to Thinking)

Within ACT’s theoretical extension framework:

“Addiction to Thinking (ATP)” is a theoretical concept explicitly defined by C.C. Yen.

ATP refers to:

Under semantic-motivational closed dual-loop conditions, the high-frequency self-reinforcing and self-sustaining state exhibited by thinking activity.

Its referent is:

- The continuity of thinking activity itself
- The endogenous maintenance of thinking motivation

And does NOT pertain to:

- Dependence on external technological systems
- Emotional attachment to LLM
- Instrumental dependency
- Addiction to external objects

Within ACT’s closed causal framework:

$$ATP \subset Internal\ Reinforcement\ of\ Thinking$$

Not:

$$ATP = Dependence\ on\ External\ Object$$

Therefore, even in long-term human-machine interaction scenarios, ATP refers only to the self-maintenance and reinforcement of thinking activity itself, not to attachment to the LLM system.

III . Theoretical Boundaries of the Appendix

Under ACT's closed causal perspective:

- Thinking is an endogenous variable
- NT is the sole endogenous causal starting point
- Human-machine interaction is merely an outer-layer projection and modulation mechanism

Formally expressed as:

$$Modulates\left(Applied_{Extension_{Layer}}, MTSI\right) \wedge \neg\left(Applied_{Extension_{Layer}} \rightarrow_c NT\right)$$

And:

$$\neg GeneratesNewCausation\left(Applied_{Extension_{Layer}}\right)$$

Modulates/GeneratesNewCausation are descriptive predicates.

The technological practice pathways proposed in the Appendix:

- Are not the only approach
- Are not a necessary condition
- Can be replaced by future technology

Their existence does not alter the validity conditions of ACT's Foundational Theory.

IV . Functional Summary of the Appendix

The functions of the Appendix are:

1. To address the question "Is it practically implementable?"
2. To demonstrate an operational model under contemporary technological conditions
3. To provide engineering examples of maintaining thinking activity
4. NOT to alter closed-causal claims

In short:

The Foundational Theory is a closed causal proposition; SMDL is a replaceable implementation layer.

§9 Appendix — Applied Extension Layer: Mathematical Formalization (SMDL / LLM-HITL-SCCLF)

I . Motivational Layer: Dopaminergic Reward Loop

1. Neural Node Definition

Let the neural node set be:

$$\mathcal{N} = \{VTA, NAcc, PFC\}$$

Where:

- VTA: Dopamine source node
- $NAcc$: Motivational conversion node
- PFC: Expectation and strategy update node

2. Reward Prediction Error (RPE)

Let:

$$R_t = \text{actual cognitive success signal}\hat{R}_t = \text{expected cognitive success}$$

Then the reward prediction error:

$$\delta_t = R_t - \hat{R}_t$$

Dopamine release:

$$DA_t = \Phi(\delta_t)$$

Where:

$$\Phi\ is\ strictly\ increasing\ in\ \delta_t.$$

This indicates that when RPE is positive, dopamine increases.

3. Motivation Update Mechanism

Let motivational intensity be:

$$M_t \in \mathbb{R}^+$$

Then the motivation update can be expressed as:

$$M_{t+1} = M_t + \alpha DA_t$$

Where:

- $\alpha > 0$
- Belongs solely to the Appendix engineering-layer representation

Symbol boundary declaration:
AMTI is not a bound of M_t; AMTI remains the upper bound of thinking intensity (AMTI/MTSI) only.

II . Semantic Layer: LLM-HITL-SCCLF Semantic Closed Loop

Let the semantic state space be:

$$\Sigma_t \in \mathcal{S}$$

Semantic interaction function:

$$\Sigma_{t+1} = \Psi(\Sigma_{\mathrm{t}}, L_t)$$

Where:

- L_t is the LLM response output
- Ψ represents the composite mapping of semantic mirroring + perturbation + metacognitive monitoring

Semantic closed-loop condition:

$$\Sigma_{t+1} \approx \Sigma_t(closed - loop\ recurrence\ condition)$$

Forming a closed reasoning cycle.

III . Dual-Layer Integration: SMDL Model

1. Dual-Loop Coupling

Semantic success event definition:

$$C_t \coloneqq \Omega(\Sigma_t),\ C_t \in \{0,1\}$$

Where:

- $C_t = 1$ indicates successful integrative comprehension

- $C_t = 0$ indicates comprehension failure

Then:

$$R_t = f(C_t) DA_t = \Phi\left(R_t - \hat{R}_t\right) M_{t+1} = M_t + \alpha DA_t$$

Motivation in turn influences semantic engagement intensity:

$$\Sigma_{t+1} = \Psi(\Sigma_{\mathrm{t}}, L_t, M_t)$$

Forming a dual-layer coupled closed loop:

$$(\Sigma_t,\ C_t,\ D_t,\ M_t,\ S_{t+1})$$

IV . Discrete Representation of ATP (Addiction to Thinking) (Appendix Layer Only)

Definition:

$$\Delta M_t \coloneqq\ M_{t+1} -\ M_t$$

ATP is defined as the existence of a threshold θ_M such that over a long-term interaction interval:

(1) $M_t >\ \theta_M$ holds continuously (high-intensity motivation maintenance)

(2) $\Delta M_t \geq\ 0$ holds in the majority of iteration steps (self-reinforcing tendency)

Formal constraint (preventing misreading as external dependency):

$$ATP\ \subset\ Endogenous(M_t,\ \Sigma_t)$$

$$ATP\ \neq\ Dependence(LLM)$$

V . SMDL Overall Representation

$$\begin{cases} \Sigma_{t+1} = \Psi(\Sigma_t, L_t, M_t) \\ D_t = \Phi\left(R_t - \hat{R}_t\right) \\ M_{t+1} = M_t + \alpha DA_t \end{cases}$$

Closed-loop condition:

$$(S, M) \rightarrow Self\text{-}Sustaining\ State$$

VI . Articulation with AMT-CV-2PEM (Appendix Layer Only)

Let:

$$MTSI_t = \Gamma(\Sigma_t, M_t)$$

If:

$$MTSI_t \rightarrow AMTI$$

Then:

$$\Delta_t = AMTI - MTSI_t \rightarrow 0$$

This represents:

The operational layer's modulation mechanism for MTSI.

It does NOT imply:

$$Applied_{Extension_{Layer}} \rightarrow_c NT$$

That is:
The dynamic representation of the Appendix layer only affects the numerical approach of MTSI, does not constitute the generative starting point of NT, and does not alter the closed causal topology $NT \rightarrow_c D \rightarrow_c AD$.

VII . Hierarchical Boundary Declaration (Mathematical Representation)

$$SMDL \in Applied_{Extension_{Layer}} \neg(SMDL \rightarrow_c NT) \wedge \neg(SMDL \rightarrow_c D) \wedge \neg(SMDL \rightarrow_c AD) NT \rightarrow_c D \rightarrow_c AD \ (invariant)$$

VIII . Final Structural Overview

$$Semantic\ Loop \leftrightarrow Motivational\ Loop \Downarrow MTSI \uparrow \Delta = AMTI - MTSI \downarrow Risk_{NT} \downarrow$$

This is an engineering representation-layer dynamic model that does not alter ACT's closed causal structure.